Department of Computer Science and Engineering

University of Barishal

Mid-Term Exam - 2 (8 Oct 2024)

Course: Automata Theory Marks: 50 Time: 45 Min

Instructor: Dr. Md Manjur Ahmed , Associate Professor

Questions:

1)	Define push down automata with example.	(5 Marks)
2)	Give pushdown automata that recognize the following languages: (a) $A = \{ w \in \{0, 1\}^* \mid w$ contains at least three 1s $\}$ (b) $B = \{ w \in \{0, 1\}^* \mid w = w^R$ and the length of w is odd $\}$	(8 Marks)
3)	Use the pumping lemma to prove that the language $A = \{ 0^{2n} 1^{3n} 0^n \mid n \geq$ 0 $\}$ is not context free.	(10 Marks)
4)	Prove that if we add a finite set of strings to a regular language, the result is a regular language.	(8 Marks)
5)	Write the closure properties of regular languages.	(5 Marks)
6)	Does a push down automata have memory? Justify.	(4 Marks)

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